GENERAL ENGINEERING DESIGN STANDARDS & MATERIALS DOSSIER:

1. CONCRETE & FOUNDATION ENGINEERING:
   ? Substructure concrete: Grade M40/M50 Self-Compacting Concrete (SCC) with microsilica for durability against soil sulfates.
   ? Reinforcing steel: High-yield strength deformed bars Fe 550D conforming to IS 1786 with minimum 16% elongation.
2. STRUCTURAL STEELWORK SPECIFICATIONS:
   ? Structural steel hollow sections and built-up plate girders conforming to IS 2062 Grade E350BR / E410BR.
   ? All full-penetration butt welds subject to 100% Non-Destructive Testing (NDT) via Ultrasonic and Radiographic inspection.
3. SEISMIC RESISTANCE & WIND RESISTANCE:
   ? Designed for Seismic Zone III with Zone Factor Z = 0.16 and Importance Factor I = 1.5 per IS 1893 (Part 1).
   ? Basic wind speed Vb = 50 m/s with terrain category 2 and cyclonic importance factor k4 = 1.15 per IS 875 (Part 3).

| Structural Element | Concrete Grade | Steel Grade | Cover mm | Exposure Class |
|---|---|---|---|---|
| Raft Foundation Slab | M50 Self-Compacting | Fe 550D TMT | 75 mm | Severe (Soil Contact) |
| Substructure Retaining Walls | M45 Waterproofed | Fe 550D TMT | 50 mm | Severe (Soil Contact) |
| Superstructure Mega Columns | M60 High Strength | Fe 550D TMT | 40 mm | Moderate |
| Post-Tensioned Floor Beams | M50 Prestressed | Fe 550D + Low Relax Strands | 35 mm | Mild (Internal Air-con) |
| Iconic Spire Tower Pylons | M60 Composite Core | Grade E410 Steel + Fe 550D | 50 mm | Very Severe (Wind/Weather) |

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| AMARAVATI GOVERNMENT COMPLEX · LEGISLATIVE ASSEMBLY | | |
| DWG TITLE: AMARAVATI GOVERNMENT COMPLEX GENERAL ENGINEER | | |
| DRAWING NO: Design_Specification | REV: R1 | SCALE: 1:100 |
| CLIENT: AP CRDA / GOVT OF ANDHRA PRADESH | DATED: 28-09-2026 | |